HAC: Node-Level Hybrid Force–Graph Control and Smooth Curriculum Scheduling for Resilient UAV Swarm Connectivity Recovery

Abstract

Self-healing communication connectivity (SCC) for unmanned aerial vehicle (UAV) swarms requires rapidly reconnecting fractured subnetworks after partial node failures while preserving wide spatial coverage, yet pursuing either objective in isolation directly degrades the other. Recent hybrid frameworks such as R2C address this tension by combining intra-subnetwork virtual-force expansion with subnetwork-level multipartite graph convolution under a steep time-adaptive fusion schedule, which bundles together several unisolated architectural choices. In this work, we formulate a three-axis taxonomy of the SCC design space spanning decision paradigm, spatial reasoning granularity, and trade-off scheduling, and systematically isolate each axis through controlled multi-seed experiments. We propose Hybrid Adaptive Convolution (HAC), a lightweight node-level hybrid controller that fuses a parameter-free buffered virtual-force heuristic with a single global graph convolutional policy at the velocity level without subnetwork partitioning, convex-hull boundary extraction, or multipartite graph construction. Across random and uniform deployment scenarios under up to 80% node failure, HAC under a constant 50/50 fusion weight outperforms purely heuristic, purely learned, and subnetwork-level rigid-translation baselines with substantially lower cross-seed variance, while retaining 88.7% to 90.7% of the coverage reported by R2C (6.2 to 7.7 pp absolute difference) under 80% node loss. Furthermore, we demonstrate that transplanting steep adaptive fusion schedules into simplified hybrid controllers degrades recovery robustness, whereas widening the transition window restores near-optimal recovery performance. Both theoretical analysis and granularity-substitution counter-experiments reveal that the fusion schedule acts as an implicit training curriculum governed by autonomy exposure, which stabilizes node-level policies in a residual-correction regime while failing to rescue subnetwork-level architectures whose rigid output spaces lack node-level expressiveness.

1 Introduction

1.1 Problem Context and Fundamental Trade-Off

Unmanned aerial vehicle (UAV) swarms deployed in contested or disaster-struck environments face recurring partial node failures that fracture communication topologies and degrade spatial coverage. Wireless aerial networks provide rapid communication coverage in post-disaster relief, dynamic battlefields, and congested areas (Coletta et al., 2023; Shi et al., 2025; Wang et al., 2022, 2024b). During operation, internal hardware faults or external attacks can disable a subset of UAVs at any time (Wang et al., 2024a; Kurunathan et al., 2023). When substantial node loss occurs, the surviving communication graph splits into multiple disconnected components, severely disrupting both multi-hop data routing and mission-area coverage (Li et al., 2023; Peng et al., 2022). The self-healing communication connectivity (SCC) problem seeks a velocity control policy that repositions surviving UAVs so that the communication graph regains full

connectivity as rapidly as possible while preserving maximal spatial coverage across the mission area.

Accelerating topological reconnection and preserving spatial coverage impose directly opposing kinematic demands on surviving agents. Pulling surviving UAVs inward toward a common center shortens reconnection time but contracts the covered footprint, whereas dispersing UAVs outward preserves area coverage at the risk of slowing or preventing link restoration. Managing this tension becomes increasingly difficult as the failure ratio approaches extreme regimes where surviving nodes are sparsely scattered across the arena.

1.2 Limitations of Existing Paradigms

Purely heuristic recovery methods require no training data but scale poorly under severe, unstructured node failures. Classical non-learning approaches either identify critical cut-vertices and trigger cascaded neighbor substitutions (Akkaya et al., 2009; Mi and Yang, 2015; Zhang et al., 2024) or guide surviving agents toward predefined rendezvous points using artificial potential fields (Joshi, 2012; Shriwastav, 2018; Chen et al., 2020; Wang et al., 2024c). Because these methods rely on static topological assumptions and local interaction rules, they become brittle when large-scale failures isolate components beyond local sensing ranges (Tosun et al., 2024).

Purely learned graph neural network (GNN) policies achieve rapid reconnection but suffer from spatial collapse. Learning-based methods formulate SCC as a differentiable graph optimization or reinforcement learning problem (Mou et al., 2022; Peng et al., 2025; Wang et al., 2025b). While GNN policies adapt effectively to diverse failure topologies, neighborhood message aggregation biases surviving UAVs to congregate around high-degree hubs, causing severe spatial clustering and coverage loss. Diffusion-based and differential-subgraph convolutions (Lin et al., 2024; Lin and Ding, 2025) partially alleviate clustering by promoting multi-hop dispersion, yet they struggle to maintain stable layouts under extreme fragmentation and rarely enforce inter-agent spacing guarantees.

1.3 Disentangling Hybrid Recovery Architecture and Scheduling

Recent divide-and-conquer hybrid frameworks achieve strong performance but bundle spatial granularity with steep adaptive scheduling. To reconcile connectivity and coverage, Peng et al. (2026) introduced R2C, which combines a buffered virtual-force module for intra-subnetwork expansion (ISE) with a multipartite graph convolution module for inter-subnetwork translation (IST) governed by an adaptive weighted fusion (AWF) schedule. Because R2C's ablations toggle entire modules on or off, two fundamental design questions remain open: whether the learned branch must operate at subnetwork granularity with explicit convex-hull boundary extraction rather than globally at the node level, and whether adaptive fusion requires a steep temporal transition rather than a smooth or constant weighting.

Controlled multi-axis ablations reveal that node-level hybrid fusion with fixed or smooth scheduling captures the vast majority of recovery performance without geometric partitioning. We investigate these questions through Hybrid Adaptive Convolution (HAC), a streamlined architecture that fuses a parameter-free buffered virtual-force branch with a single global graph convolutional policy at the velocity level. Our systematic evaluation yields four principal findings:

1. Fusing a global node-level learned branch with a parameter-free virtual-force heuristic under a constant 50/50 weight (HAC-Fixed) retains 88.7% to 90.7% of R2C's reported coverage under 80% node failure while achieving near-perfect recovery rates and 3× to 5× lower cross-seed

variance than single-branch baselines.
2. Directly applying a steep adaptive fusion schedule to this simplified architecture severely degrades recovery robustness under heavy node loss, reversing the trend observed in complex multipartite architectures.
3. Recovery degradation under adaptive gating is governed by transition steepness rather than transition timing, as widening the sigmoid transition band restores near-optimal success rates by providing an implicit residual-correction training curriculum.
4. Schedule smoothing rescues node-level hybrid control (HAC) but fails to rescue subnetwork-level rigid-translation hybrid control (HSC), cleanly separating a training-schedule bottleneck from an architectural expressiveness bottleneck.

1.4 Summary of Contributions

The primary contributions of this paper are fourfold:

1. We formulate a three-axis taxonomy of the SCC design space spanning decision paradigm, spatial reasoning granularity, and trade-off strategy, positioning thirteen representative methods and isolating previously unexplored hybrid configurations.
2. We propose HAC, a lightweight hybrid recovery architecture aided by a cardinality-invariant soft-minimum connectivity surrogate loss, and validate its resilience and variance-suppression properties across random and uniform deployments over five independent random seeds.
3. We establish a theoretical and empirical account of hybrid fusion scheduling based on link-retention barriers and training-time autonomy exposure, deriving a falsifiable prediction that relates transition slope to closed-loop learning stability.
4. We provide a complete comparative analysis separating node-level versus subnetwork-level hybrid expressiveness, accompanied by open-source simulation code, training scripts, and multi-seed evaluation logs.

The remainder of this paper is organized as follows. Section 2 reviews related work, and Section 3 formalizes the SCC problem and design taxonomy. Section 4 presents motivating experiments on single-branch failure modes. Section 5 details the HAC architecture and cardinality-invariant loss design, followed by theoretical analysis in Section 6. Sections 7 through 9 report the experimental setup, main benchmarks, and ablation studies. Section 10 discusses design implications and limitations, and Section 11 concludes the paper.

2 Related Work

2.1 Non-Learning Self-Healing Methods

Critical-node compensation algorithms restore local links through cascaded neighbor movements but incur prohibitive coordination overhead as swarm scale grows. Early self-healing protocols detect cut-vertices whose removal partitions the network and trigger sequential backup replacements (Abbasi and Younis, 2008; Akkaya et al., 2009; Mi and Yang, 2015; Zhang et al., 2024). While effective for isolated single-node faults, identifying critical nodes and coordinating multi-hop cascades scale poorly when simultaneous failures shatter the topology into numerous disjoint clusters.

Rendezvous-point aggregation methods eliminate cut-vertex search but rely on static geometric rules that falter under unpredictable failure patterns. Alternative heuristic schemes direct surviving UAVs or fractured subnetworks toward predetermined landmarks using

Coulomb-inspired forces (Joshi, 2012), round-table negotiation (Shriwastav, 2018), or artificial potential fields (Chen et al., 2020; Wang et al., 2024c). Although parameter-free and computationally lightweight, these hand-crafted rules lack global topological awareness and frequently stall in local equilibria or cause excessive coverage contraction under high failure ratios (Tosun et al., 2024).

2.2 Learning-Based Self-Healing Methods

Graph neural network policies accelerate global reconnection but exhibit severe spatial collapse around high-degree nodes. Posing connectivity recovery as a differentiable graph optimization problem enables end-to-end learning of coordinated trajectories (Karalias and Loukas, 2020; Peng et al., 2025). Mou et al. (2022) employ a relaxed GCN over virtual communication links, while Guo et al. (2024) incorporate spatiotemporal multi-head attention. Despite fast convergence, standard message-passing schemes pull peripheral agents toward dense clusters, degrading coverage uniformity.

Diffusion and differential-subgraph architectures improve spatial dispersion yet remain vulnerable to extreme topological fragmentation. Recent works counter spatial collapse by introducing multi-hop diffusion convolutions (Lin et al., 2024) or bipartite message passing over differential subgraphs (Lin and Ding, 2025). While these mechanisms encourage broader spatial spread, purely learned policies lack hard repulsive buffers to guarantee inter-agent spacing and exhibit high sensitivity to weight initialization when up to 80% of the swarm is lost.

2.3 Hybrid Heuristic–Learning Architectures and Positioning

Hybrid recovery frameworks combine physical priors with learned graph representations to balance local spacing and global bridging. Peng et al. (2026) proposed R2C, which pairs a buffered Hookean force field for intra-subnetwork expansion (ISE) with a multipartite spectral GCN for inter-subnetwork translation (IST) using Quickhull boundary extraction (Barber et al., 1996) and asymmetric spectral filtering (Liu and Wang, 2025), blended via a sigmoid adaptive weighted fusion (AWF) schedule (Elfwing et al., 2018). Elsewhere, Li et al. (2025) combine distributed multi-agent reinforcement learning with artificial potential fields at the node level under fixed weighting, while Wang et al. (2025b) and related multi-robot coordination studies (Ye et al., 2022; Mutzari et al., 2025; Shi et al., 2025; Kipf and Welling, 2017; Zeng et al., 2023) explore graph-based control without adaptive heuristic-learned velocity fusion.

Our work isolates spatial reasoning granularity and schedule smoothness as independent design dimensions within the hybrid recovery paradigm. Whereas prior studies evaluate hybrid modules as monolithic packages, we decompose the SCC design space along decision paradigm, spatial granularity, and trade-off scheduling. By holding the training objective and graph backbone fixed while systematically varying spatial granularity (node-level global vs. subnetwork-level rigid translation) and fusion slope, we identify the minimal architectural and scheduling conditions required for resilient recovery.

3 Preliminaries and Problem Formulation

3.1 UAV Swarm Network Model

The surviving UAV swarm is modeled as a time-varying proximity graph over a bounded planar mission area. Following Peng et al. (2026), at discrete time step t, the swarm forms a

temporal graph $G_t = (V_t, E_t, P_t)$, where $V_t = \{v_{1,t}, \ldots, v_{|V_t|,t}\}$ denotes the surviving UAVs, E_t denotes the active communication links, and $P_t = \{\mathbf{p}_{1,t}, \ldots, \mathbf{p}_{|V_t|,t}\}$ with $\mathbf{p}_{i,t} \in \mathcal{A} \subset \mathbb{R}^2$ represents their planar coordinates. Let $\mathbf{A}_t$ be the binary adjacency matrix and $\mathbf{L}_t = \mathbf{D}_t - \mathbf{A}_t$ be the graph Laplacian. The number of connected components C_t equals the algebraic multiplicity of the zero eigenvalue of $\mathbf{L}_t$, and global connectivity holds if and only if $C_t = 1$.

Communication links and ground coverage follow Euclidean disk models governed by air-to-air path loss. An undirected link $(v_i, v_j) \in E_t$ exists if and only if the inter-UAV distance $d_{ij,t} = \|\mathbf{p}_{i,t} - \mathbf{p}_{j,t}\|_2$ satisfies $d_{ij,t} \leq R_c$, where R_c is the communication range derived from free-space air-to-air channel models (Ma et al., 2019; Ni et al., 2024; Wang et al., 2025a). Each UAV covers a circular disk $\mathcal{C}_i(t)$ of radius R_{cov}, yielding a collective swarm coverage ratio $C_{\text{cov}}(t) = \left|\bigcup_{v_i \in V_t} \mathcal{C}_i(t)\right| / |\mathcal{A}|$.

3.2 Optimization Objective and Kinematic Constraints

The self-healing connectivity recovery objective jointly penalizes reconnection delay and post-recovery coverage loss. Let t_1 denote the failure onset step and $t_r = \min\{t \geq t_1 : C_t = 1\}$ denote the earliest step at which full connectivity is restored within horizon T, giving a normalized recovery time ratio $T_{\text{rec}} = (t_r - t_1)/T$. We seek a velocity control policy $\hat{V}_t = \{\hat{\mathbf{v}}_{1,t}, \ldots, \hat{\mathbf{v}}_{|V_t|,t}\}$ that minimizes the composite cost

$$J_c = T_{\text{rec}} + \big(1 - C_{\text{cov}}(t_r)\big), \tag{1}$$

subject to four operational constraints: (C1) terminal connectivity $C_{t_r} = 1$; (C2) speed and arena bounds $\|\hat{\mathbf{v}}_{i,t}\|_2 \leq \hat{v}_{\max}$ and $\mathbf{p}_{i,t} \in \mathcal{A}$; (C3) discrete-time Euler kinematics $\mathbf{p}_{i,t} = \mathbf{p}_{i,t_1} + \sum_{t_1 \leq k < t} \Delta t \cdot \hat{\mathbf{v}}_{i,k}$; and (C4) link feasibility $d_{ij,t} \leq R_c$ for all active intra-swarm transmissions. Minimizing T_{rec} requires inward convergence, whereas minimizing $1 - C_{\text{cov}}(t_r)$ requires spatial dispersion.

3.3 A Three-Axis Taxonomy of the Design Space

Every self-healing recovery method can be categorized along three orthogonal design axes: decision paradigm, spatial granularity, and trade-off strategy.

- **Axis 1 (Decision Paradigm):** Motion commands are generated by a *heuristic* rule with zero learnable parameters, a *learned* parametric network trained via data or differentiable simulation, or a *hybrid* controller that explicitly fuses heuristic and learned outputs.
- **Axis 2 (Spatial Reasoning Granularity):** Control reasoning operates at the *node-level* using local neighborhood states, the *subnetwork-level* where each connected component acts as a rigid or coordinated unit, or the *global-level* where all surviving agents exchange context across the entire arena.
- **Axis 3 (Connectivity–Coverage Trade-Off Strategy):** Competing objectives are balanced via a *fixed* constant weight, a *staged* sequential switch, or a continuous *adaptive* schedule parameterized by time or topological state.

Table 1 maps representative prior methods and our four experimental architectures (GAC, SAC, HAC, and HSC) into this $3 \times 3 \times 3$ combinatorial space.

4 Motivating Experiments: Failure Modes of Single-Branch Designs

To motivate our hybrid architecture, we first isolate two purely learned baselines that share identical loss functions and network backbones but operate at opposite spatial granularities: node-level global

Table 1: Positioning of prior SCC methods and the architectures evaluated in this work within the $3 \times 3 \times 3$ design taxonomy.

Paradigm	Granularity	Trade-off	Representative method(s)
Heuristic	Node-level	Fixed	Akkaya et al. (2009); Mi and Yang (2015); Wang et al. (2024c)
Heuristic	Subnetwork	Fixed	Chen et al. (2020)
Heuristic	Global	Fixed	Lin and Ding (2025)
Heuristic	Global	Staged	HERO (Mi et al., 2011)
Learned	Node-level	Fixed	Mou et al. (2022); **GAC** (Section 4)
Learned	Subnetwork	Fixed	Lin et al. (2024); **SAC** (Section 4)
Learned	Global	Fixed	Guo et al. (2024)
Hybrid	Node-level	Fixed	Li et al. (2025)
Hybrid	Node+Subnet	Adaptive	R2C (ISE+IST+AWF) (Peng et al., 2026)
Hybrid	**Global**	**Fixed / Adaptive**	**HAC (this work, Section 5)**
Hybrid	**Subnetwork**	**Fixed / Adaptive**	**HSC (this work, Section 9)**

convolution (GAC) and subnetwork-level rigid translation (SAC).

4.1 Global-Level Learned Policy (GAC)

A purely learned global graph policy (GAC) achieves moderate reconnection capability but suffers from severe spatial collapse. GAC constructs a k-nearest-neighbor (k-NN) graph over the Euclidean coordinates of all surviving UAVs regardless of communication range R_c, allowing message passing across disconnected components. A 4-layer GCN with hidden dimension 128 and a two-layer MLP head directly outputs per-node 2D velocities scaled via tanh by $\hat{v}_{\max}$ (without the unit-norm velocity projection used in hybrid fusion), trained end-to-end with equal weights on connectivity and coverage surrogate losses. As shown in Table 2, GAC's recovery success rate drops from 49.0 ± 0.6 out of 50 trials at 20% failure to 19.8 ± 5.5 at 80% failure, accompanied by an elevated average node degree (AND = 13.68 at 20% failure) that reflects severe spatial clustering around high-degree nodes.

4.2 Subnetwork-Level Rigid Translation (SAC)

Restricting the learned policy to subnetwork-level rigid translation (SAC) prevents spatial clustering but causes deterministic recovery collapse under high failure rates. SAC partitions surviving UAVs into connected components under R_c, pools each component into its centroid position and relative size, and applies the same 4-layer GCN over a k-NN graph of component centroids to predict a single rigid 2D translation vector shared by all members of each subnetwork. Table 2 demonstrates that SAC maintains a low, uniform node degree (AND = 6.82 at 20% failure), eliminating intra-cluster collapse. However, its recovery success rate plummets to 5.0 ± 0.0 out of 50 at 60% failure and 0.0 ± 0.0 at 80% failure across all five seeds. Rigid component translation lacks the internal degrees of freedom needed to simultaneously bridge multiple scattered singletons without tearing existing links.

Table 2: Motivating baseline comparison under random deployment (mean ± std over 5 seeds, 50 trials per seed). GAC suffers from spatial clustering (high AND), whereas SAC avoids clustering but collapses deterministically at high failure rates.

Failure	GAC (global node-level)		SAC (subnetwork rigid)	
Rate	AND	Success/50	AND	Success/50
20%	13.68 ± 0.28	49.0 ± 0.6	6.82 ± 0.00	40.0 ± 0.0
40%	12.48 ± 0.61	47.2 ± 1.7	5.85 ± 0.00	26.0 ± 0.0
60%	10.55 ± 0.78	42.8 ± 3.7	4.18 ± 0.00	5.0 ± 0.0
80%	6.00 ± 0.26	19.8 ± 5.5	n/a	0.0 ± 0.0

4.3 Complementary Failure Modes Motivate Node-Level Hybridization

The opposite failure modes of GAC and SAC demonstrate that resilient recovery requires both node-level movement flexibility and an explicit local spacing prior. GAC retains full $2|V_t|$ spatial degrees of freedom to bridge fragmented clusters but lacks a short-range repulsive mechanism to prevent over-concentration. Conversely, SAC preserves internal spacing by freezing relative intra-component positions but loses the ability to deform components toward multiple targets. This complementarity directly motivates fusing a parameter-free local virtual-force field with a node-level global GCN policy.

5 Method: Hybrid Adaptive Convolution (HAC)

5.1 Architecture Overview

Hybrid Adaptive Convolution (HAC) fuses a parameter-free buffered virtual-force heuristic with a single global graph convolutional network at the individual node velocity level. Unlike multipartite subnetwork architectures that require Quickhull boundary extraction and cross-subgraph virtual edge construction, HAC operates directly on the homogeneous set of surviving UAVs. This streamlined design isolates the core benefit of heuristic–learned velocity coupling from complex geometric preprocessing.

5.2 Heuristic Branch: Buffered Three-Zone Virtual-Force Field

The heuristic branch regulates local inter-UAV spacing and arena containment through a parameter-free three-zone potential field. Adopting the buffered virtual-spring formulation (Rychlewski, 1984; Peng et al., 2026), we partition the communication disk $(0, R_c]$ around an equilibrium buffer zone $[d_l, d_u] = [0.8R_c, 0.9R_c]$. The pairwise virtual force exerted on UAV v_i by a communicating neighbor $v_j \in \mathcal{N}_i = \{v_j \in V_t : 0 < d_{ij,t} \leq R_c\}$ is defined as

$$\mathbf{F}'_{ij,t} = \begin{cases} -\exp\left(-\alpha \cdot \dfrac{d_{ij,t}}{d_l}\right) \mathbf{u}_{ij,t}, & 0 \leq d_{ij,t} < d_l \\ \mathbf{0}, & d_l \leq d_{ij,t} \leq d_u \\ \exp\left(-\beta \cdot \dfrac{R_c - d_{ij,t}}{R_c - d_u}\right) \mathbf{u}_{ij,t}, & d_u < d_{ij,t} \leq R_c, \end{cases} \tag{2}$$

where $\mathbf{u}_{ij,t} = (\mathbf{p}_{j,t} - \mathbf{p}_{i,t})/d_{ij,t}$ is the unit direction vector from v_i to v_j, and $\alpha = \beta = 0.01$ govern the exponential decay rates in the repulsive zone $[0, d_l)$ and attractive buffer zone $(d_u, R_c]$. To

prevent agents from exiting the mission area $\mathcal{A}$, any UAV within distance $d_{i,k} < d_e = 0.15R_c$ of boundary edge $k \in \mathcal{B} = \{\text{left, right, bottom, top}\}$ experiences an inward normal repulsion $\mathbf{F}''_{i,t} = \sum_{k\in\mathcal{B}} \exp(-\delta \cdot d_{i,k})\, \mathbf{n}_k$ with $\delta = 0.01$. Summing $\mathbf{F}_{i,t} = \sum_{j\in\mathcal{N}_i} \mathbf{F}'_{ij,t} + \mathbf{F}''_{i,t}$ and normalizing yields the heuristic velocity command $\mathbf{v}'_{i,t} = \hat{v}_{\max} \cdot \mathbf{F}_{i,t}/(\|\mathbf{F}_{i,t}\|_2 + \epsilon)$. Because Eq. (2) contains no trainable weights, $\mathbf{v}'_{i,t}$ provides an invariant physical anchor throughout both training and inference.

5.3 Learned Branch and Velocity-Level Adaptive Fusion

The learned branch propagates topological context across disconnected components via a global k-NN graph convolutional network. At each step t, we construct a symmetric k-NN graph ($k = 8$) over the Euclidean coordinates of all surviving UAVs V_t. Each node v_i receives a 3-dimensional input feature vector comprising its normalized planar coordinates and its normalized k-NN degree. This feature representation passes through an $L = 4$ layer GCN with hidden dimension 128 and ReLU activations, followed by a two-layer MLP head with tanh activation scaled by $\hat{v}_{\max}$ to produce the learned velocity vector $\mathbf{v}''_{i,t}$.

Velocity-level fusion combines heuristic and learned commands through a tunable sigmoid schedule followed by unit-speed renormalization. To govern the temporal handover from heuristic expansion to learned global translation, we define the time-dependent gating weights

$$w_T(t) = \frac{1}{1 + \exp\big(-k \cdot (t/T - s_0)\big)}, \qquad w_E(t) = 1 - w_T(t), \tag{3}$$

where $k > 0$ controls transition steepness and $s_0 \in (0, 1)$ specifies the crossover midpoint relative to horizon T. The executed velocity command $\hat{\mathbf{v}}_{i,t}$ is obtained by renormalizing the weighted sum:

$$\hat{\mathbf{v}}_{i,t} = \hat{v}_{\max} \cdot \frac{w_E(t)\, \mathbf{v}'_{i,t} + w_T(t)\, \mathbf{v}''_{i,t}}{\left\| w_E(t)\, \mathbf{v}'_{i,t} + w_T(t)\, \mathbf{v}''_{i,t} \right\|_2 + \epsilon}. \tag{4}$$

We evaluate three primary configurations of Eq. (4): **HAC-Fixed**, which sets constant equal weights $w_E(t) = w_T(t) = 0.5$; **HAC-Adaptive**, which varies (k, s_0) in Eq. (3); and the **Learned-only** ablation, which sets $w_T(t) = 1$ and $w_E(t) = 0$ inside Eq. (4) so that $\mathbf{v}''_{i,t}$ undergoes exact unit-norm projection $\hat{v}_{\max}\mathbf{v}''_{i,t}/(\|\mathbf{v}''_{i,t}\|_2 + \epsilon)$ (distinguishing it from GAC, which uses unprojected tanh scaling).

5.4 Cardinality-Invariant Training Objective and Algorithms

A cardinality-invariant soft-minimum connectivity surrogate eliminates component-size bias during differentiable multi-step rollout training. Because the discrete component count C_t is non-differentiable, we supervise inter-component bridging using a smooth soft-minimum distance surrogate over every pair of currently disconnected components S_i, S_j:

$$\mathcal{L}_{\text{conn}}^{(i,j)} = -\tau \left(\log \sum_{\substack{u\in S_i \\ v\in S_j}} \exp\left(-\frac{d_{uv}}{\tau}\right) - \log(n_i n_j) \right), \tag{5}$$

averaged over all disconnected component pairs, where $n_i = |S_i|$, $n_j = |S_j|$, and $\tau > 0$ is a temperature parameter. Standard log-sum-exp soft-minimum estimators omit the $-\log(n_i n_j)$ correction, introducing an additive distortion of $-\tau \log(n_i n_j)$ that systematically under-penalizes large components and yields spuriously negative loss values (verified numerically in Appendix A). Including

Algorithm 1 HAC Differentiable Rollout Training (One Epoch)

Require: Learned policy π_θ, training rollout length T_{train}, weight schedule $w_T(\cdot)$

Sample failure ratio $r \sim \{0.2, 0.4, 0.6, 0.8\}$, initialize positions $\mathbf{P}_0$, and remove failed nodes
$\mathcal{L}_{\text{total}} \leftarrow 0$
for $t = 0, \ldots, T_{\text{train}} - 1$ **do**
    Identify connected components $\{S_m\}$ of $\mathbf{P}_t$ under communication range R_c
    Compute heuristic velocities $\mathbf{v}'$ via Eq. (2) ▷ detached, zero trainable parameters
    Construct global k-NN graph over $\mathbf{P}_t$ and evaluate $\mathbf{v}'' \leftarrow \pi_\theta(\mathbf{P}_t)$ ▷ learned branch
    Evaluate gating weights $w_T \leftarrow w_T(t)$ and $w_E \leftarrow 1 - w_T$
    Fuse velocities $\hat{\mathbf{v}} \leftarrow \text{Fuse}(\mathbf{v}', \mathbf{v}'', w_E, w_T)$ via Eq. (4)
    Update positions $\mathbf{P}_{t+1} \leftarrow \text{clip}(\mathbf{P}_t + \Delta t \cdot \hat{\mathbf{v}},\ 0,\ L)$
    $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{total}} + 0.5\,\mathcal{L}_{\text{conn}}(\mathbf{P}_{t+1}) + 0.5\,\mathcal{L}_{\text{cov}}(\mathbf{P}_{t+1}) + \lambda\,\mathcal{L}_{\text{move}}$
end for
Backpropagate $\mathcal{L}_{\text{total}}/T_{\text{train}}$ through π_θ and update θ via Adam with gradient clipping

Algorithm 2 HAC Inference Deployment Rollout

Require: Trained policy π_θ, evaluation horizon T, weight schedule $w_T(\cdot)$

for $t = 0, \ldots, T - 1$ **do**
    if $C_t = 1$ (swarm is fully connected) **then return** success, recovery step t
    end if
    Compute heuristic velocities $\mathbf{v}'$ via Eq. (2)
    Evaluate learned velocities $\mathbf{v}'' \leftarrow \pi_\theta(\mathbf{P}_t)$ ▷ no gradient computation
    Fuse velocities $\hat{\mathbf{v}} \leftarrow \text{Fuse}(\mathbf{v}', \mathbf{v}'', 1 - w_T(t), w_T(t))$ via Eq. (4)
    Update positions $\mathbf{P}_{t+1} \leftarrow \text{clip}(\mathbf{P}_t + \Delta t \cdot \hat{\mathbf{v}},\ 0,\ L)$
end for
return failure (horizon T exhausted)

$-\log(n_i n_j)$ ensures exact cardinality invariance so that isolated singletons receive gradient magnitudes consistent with large subnetworks.

Spatial coverage and movement efficiency are supervised jointly through differentiable grid-coverage and displacement regularizers. We approximate area coverage over a uniform grid $G \subset \mathcal{A}$ via $\mathcal{L}_{\text{cov}} = \frac{1}{|G|} \sum_{\mathbf{g} \in G} \big(1 - \sigma\big((R_{\text{cov}} - \min_i \|\mathbf{g} - \mathbf{p}_i\|_2)/\tau'\big)\big)$ and penalize excessive control effort via $\mathcal{L}_{\text{move}} = \frac{1}{|V_t|} \sum_i \|\mathbf{p}_{i,t+1} - \mathbf{p}_{i,t}\|_2$. Throughout training, connectivity and coverage losses receive equal static weights of 0.5, ensuring that all temporal trade-offs occur strictly at the velocity fusion layer in Eq. (4). Algorithms 1 and 2 summarize end-to-end differentiable rollout training and closed-loop deployment.

6 Theoretical Analysis

Our ablation studies in Section 9 demonstrate that applying a steep adaptive schedule ($k = 30, s_0 = 0.125$) to HAC degrades recovery success under 80% failure, whereas reducing the transition slope k restores robustness. Here we establish two complementary mechanisms explaining why transition steepness governs hybrid recovery: an inference-time link-retention barrier in the heuristic branch and a training-time implicit curriculum governed by autonomy exposure.

6.1 Link-Retention Barrier of the Heuristic Branch

Proposition 1 (Link-retention barrier in the attractive buffer zone)**.** *Let v_i, v_j be two connected UAVs occupying the attractive buffer zone $d_u < d_{ij,t} \le R_c$ of Eq. (2), and suppose no external repulsive or boundary force acts on v_i or v_j. Then, for any step size $0 < \Delta t < d_{ij,t}/(2\hat{v}_{\max})$, executing the heuristic branch alone ($w_E = 1, w_T = 0$ in Eq. (4)) strictly decreases their inter-UAV distance according to $d_{ij,t+\Delta t} = d_{ij,t} - 2\Delta t\,\hat{v}_{\max} < R_c$, thereby guaranteeing that the existing communication link $(v_i, v_j) \in E_t$ cannot break at step $t + \Delta t$.*

Proof. By Eq. (2), the pairwise force on v_i from v_j within $d_u < d_{ij,t} \le R_c$ is $\mathbf{F}'_{ij,t} = \exp\!\big(-\beta \frac{R_c - d_{ij,t}}{R_c - d_u}\big)\mathbf{u}_{ij,t}$ pointing from v_i toward v_j. By symmetry, $\mathbf{F}'_{ji,t} = -\mathbf{F}'_{ij,t}$ points from v_j toward v_i with equal magnitude. Under the isolation assumption, $\mathbf{F}_{i,t} = \mathbf{F}'_{ij,t}$ and $\mathbf{F}_{j,t} = -\mathbf{F}'_{ij,t}$, which after unit-speed normalization yields $\mathbf{v}'_{i,t} = \hat{v}_{\max}\,\mathbf{u}_{ij,t}$ and $\mathbf{v}'_{j,t} = -\hat{v}_{\max}\,\mathbf{u}_{ij,t}$. Applying the one-step Euler update gives $d_{ij,t+\Delta t} = \|\mathbf{p}_{i,t} + \Delta t\,\mathbf{v}'_{i,t} - \mathbf{p}_{j,t} - \Delta t\,\mathbf{v}'_{j,t}\|_2 = \|(\mathbf{p}_{i,t} - \mathbf{p}_{j,t}) + 2\Delta t\,\hat{v}_{\max}\,\mathbf{u}_{ij,t}\|_2$. Substituting $\mathbf{p}_{i,t} - \mathbf{p}_{j,t} = -d_{ij,t}\,\mathbf{u}_{ij,t}$ yields $d_{ij,t+\Delta t} = |d_{ij,t} - 2\Delta t\,\hat{v}_{\max}| = d_{ij,t} - 2\Delta t\,\hat{v}_{\max} < d_{ij,t} \le R_c$ for all $0 < \Delta t < d_{ij,t}/(2\hat{v}_{\max})$. □

Remark 1 (Complementary roles of intra-subnetwork zones and handover risk)**.** *Proposition 1 establishes that the attractive buffer zone $(d_u, R_c]$ acts as a local restorative barrier that prevents existing communication links from fracturing, while the repulsive zone $[0, d_l)$ drives intra-subnetwork expansion to push boundary nodes outward toward disconnected components. Because the learned branch $\mathbf{v}''_{i,t}$ carries no structural barrier guarantee, prematurely suppressing $w_E(t)$ below a critical margin allows imperfect learned velocities to violate link-retention invariance and tear newly formed bridges.*

6.2 Fusion Weight as an Implicit Training Curriculum

During end-to-end differentiable rollout training, the fusion schedule acts as an implicit curriculum that determines whether the learned branch masters residual correction before attempting autonomous control. Because π_θ is trained directly through the fused velocity Eq. (4) in Algorithm 1, the temporal profile of $w_T(t)$ shapes the supervision landscape at every rollout step.

Definition 1 (Autonomy exposure)**.** *For a training rollout of length T_{train} and dominance threshold $\tau \in (0, 1)$, the* autonomy exposure *of schedule $w_T(\cdot; k, s_0)$ is the fraction of training steps during which the learned branch holds majority control:*

$$\mathrm{AE}(k, s_0; \tau) = \frac{1}{T_{\text{train}}} \Big| \big\{\, t \in \{0, \dots, T_{\text{train}} - 1\} : w_T(t; k, s_0) > \tau \,\big\} \Big|. \tag{6}$$

For $\tau = 0.5$, inverting Eq. (3) between $w_T = 0.1$ and $w_T = 0.9$ yields a transition window of width $\Delta t \approx \frac{\ln(81)}{k}\, T_{\text{train}} \approx \frac{4.394}{k}\, T_{\text{train}}$. Consequently, a steep slope ($k = 30$) compresses the handover into a narrow band and forces the randomly initialized GCN to operate nearly autonomously ($w_T \approx 1$) across most of the rollout, compounding open-loop trajectory errors. Conversely, a gentle slope ($k = 5$, spanning $\approx 88\%$ of the 40-step training rollout) maintains substantial heuristic guidance ($w_E > 0$) throughout training, enabling the GCN to learn stable residual corrections on top of a well-behaved physical baseline.

Remark 2 (Boundary condition of curriculum smoothing)**.** *The implicit-curriculum mechanism predicts two empirical phenomena verified in Section 9: first, smoothing the transition slope ($k = 5, s_0 = 0.7$) restores* HAC*'s recovery rate under 80% failure to match* HAC-Fixed*; second, schedule*

*smoothing cannot rescue the subnetwork-level hybrid (*HSC*), because no training curriculum can overcome the representational inability of rigid component translation to express individual node deformations. We note that single-seed intermediate points ($k = 10, s_0 = 0.125$ vs. $k = 30, s_0 = 0.125$) exhibit minor non-monotonic fluctuations within the ± 5 to 6 trial variance typical of single-seed sweeps.*

6.3 A Falsifiable Prediction for Multi-Seed Schedule Design

The autonomy-exposure formulation yields a concrete, falsifiable prediction: schedules with distinct (k, s_0) pairs but matched autonomy exposure $\mathrm{AE}(k, s_0; 0.5)$ will achieve statistically equivalent recovery robustness. Testing this invariance across multi-seed contours of constant AE provides a direct diagnostic for future curriculum-scheduled hybrid controllers.

7 Experimental Setup

7.1 Simulation Environment and Failure Protocol

All architectures are evaluated across random and uniform swarm deployments under four node failure ratios up to 80%. Mirroring the benchmark protocol of Peng et al. (2026), we simulate a $1000 \times 1000\,\mathrm{m}^2$ mission area under two initial layouts: **random deployment** with 200 UAVs sampled uniformly at random, and **uniform deployment** with 196 UAVs initialized on a regular 14×14 grid. Each UAV operates with communication and coverage radii $R_c = R_{\mathrm{cov}} = 120\,\mathrm{m}$, maximum speed $\hat{v}_{\max} = 1\,\mathrm{m/s}$, time step $\Delta t = 1\,\mathrm{s}$, and an evaluation horizon of $T = 400$ steps. We evaluate four failure ratios $r \in \{20\%, 40\%, 60\%, 80\%\}$ via uniform random node removal. For every main configuration, we train 5 independent random seeds and evaluate each seed across 50 independent failure trials, reporting the cross-seed mean and standard deviation.

7.2 Baselines and Evaluation Metrics

We compare HAC against purely heuristic, purely learned, subnetwork-level rigid, and published multipartite baselines across four topological and coverage metrics. Our controlled baselines comprise: (i) **Heuristic-only**, executing Eq. (2) in isolation ($w_E = 1, w_T = 0$); (ii) **GAC-Fixed** and **SAC-Fixed** from Section 4; (iii) **Learned-only**, setting $w_T = 1, w_E = 0$ in Eq. (4); and (iv) **HSC**, substituting SAC's subnetwork-level rigid-translation GCN into the hybrid fusion loop. We also include the published metrics of **R2C** (Peng et al., 2026) as an external reference point. Performance is quantified via four metrics:

- **Success/50:** Number of evaluation trials out of 50 that restore full connectivity ($C_t = 1$) within $T = 400$ steps. Following standard protocol, non-recovering trials are excluded from conditional post-recovery averages.
- **REC (↓):** Normalized recovery time ratio $T_{\mathrm{rec}} = (t_r - t_1)/T$ averaged over successful trials.
- **COV (↑):** Swarm coverage ratio $C_{\mathrm{cov}}(t_r)$ at the instant of reconnection, evaluated via 5 m grid sampling.
- **AND:** Average node degree at t_r, serving as an inverse proxy for spatial dispersion (elevated AND indicates spatial clustering).

7.3 Implementation and Seeding Protocol

All learned models share an identical 4-layer GCN backbone and strict full-pipeline random seeding. Every graph convolutional policy uses $L = 4$ layers, hidden width 128, $k = 8$ nearest

Table 3: Main benchmark results under random deployment (mean ± std over 5 seeds, 50 trials per seed). Here n/a denotes zero successful trials.

Architecture	Failure	Success/50 ↑	REC ↓	COV ↑
HAC-Fixed	20%	50.0 ± 0.0	0.061 ± 0.006	0.997 ± 0.000
	40%	50.0 ± 0.0	0.136 ± 0.006	0.988 ± 0.001
	60%	50.0 ± 0.0	0.307 ± 0.006	0.923 ± 0.005
	80%	$\mathbf{48.8 \pm 0.98}$	$\mathbf{0.666 \pm 0.032}$	$\mathbf{0.604 \pm 0.005}$
GAC-Fixed	20%	49.0 ± 0.6	0.324 ± 0.007	0.747 ± 0.009
	40%	47.2 ± 1.7	0.467 ± 0.020	0.614 ± 0.021
	60%	42.8 ± 3.7	0.612 ± 0.027	0.452 ± 0.022
	80%	19.8 ± 5.5	0.775 ± 0.017	0.310 ± 0.009
SAC-Fixed	20%	40.0 ± 0.0	0.216 ± 0.000	0.910 ± 0.000
	40%	26.0 ± 0.0	0.567 ± 0.000	0.713 ± 0.000
	60%	5.0 ± 0.0	0.852 ± 0.000	0.531 ± 0.000
	80%	0.0 ± 0.0	n/a	n/a
Learned-only	20%	44.4 ± 1.4	0.277 ± 0.013	0.851 ± 0.008
	40%	37.2 ± 1.6	0.402 ± 0.009	0.759 ± 0.008
	60%	31.4 ± 3.2	0.613 ± 0.021	0.558 ± 0.019
	80%	12.0 ± 3.5	0.874 ± 0.025	0.306 ± 0.016
Heuristic-only	20%	47.4 ± 1.4	0.051 ± 0.017	0.997 ± 0.000
	40%	30.4 ± 2.2	0.120 ± 0.027	0.993 ± 0.001
	60%	1.4 ± 1.4	0.224 ± 0.091	0.964 ± 0.008
	80%	0.0 ± 0.0	n/a	n/a

neighbors, movement cost weight $\lambda = 0.01$, and Adam optimization with learning rate 10^{-3} and gradient clipping at 1.0 over 300 epochs of 40-step differentiable rollouts. Both environment randomness and PyTorch weight initialization (`torch.manual_seed` and `torch.cuda.manual_seed_all`) are strictly locked per seed prior to model instantiation.

8 Results

8.1 Main Results: HAC vs. Single-Branch Baselines

HAC-Fixed achieves near-perfect connectivity recovery and superior coverage retention across both random and uniform deployments, outperforming all single-branch baselines by wide margins under 80% **node failure.** Tables 3 and 4 report the five-seed benchmark results for random and uniform deployments, and Figure 1 illustrates the success-rate trajectories under random deployment. Under 20% to 60% node loss, HAC-Fixed achieves a 100% recovery rate (50.0 ± 0.0 out of 50) in both scenarios. Under extreme 80% failure, HAC-Fixed succeeds in 48.8 ± 0.98 trials (random) and 49.0 ± 2.0 trials (uniform), outperforming the strongest single-branch baseline (GAC-Fixed at 19.8 ± 5.5 and 25.8 ± 6.1) by 29.0 and 23.2 successful trials, respectively, while nearly doubling spatial coverage (COV of 0.604 vs. 0.310 in random deployment, an improvement of 29.4 pp). Both SAC-Fixed and Heuristic-only collapse deterministically to 0.0 ± 0.0 successes at 80% failure across all seeds.

8.2 Comparison with Multipartite Subnetwork Architectures

HAC-Fixed retains 88.7% **to** 90.7% **of the coverage reported by R2C under** 80% **node failure despite omitting convex-hull boundary extraction, virtual multipartite edges, and adaptive schedule tuning.** Table 5 compares HAC-Fixed against the published figures of R2C (Peng et al., 2026). At 20% and 40% failure rates, both methods achieve virtually identical

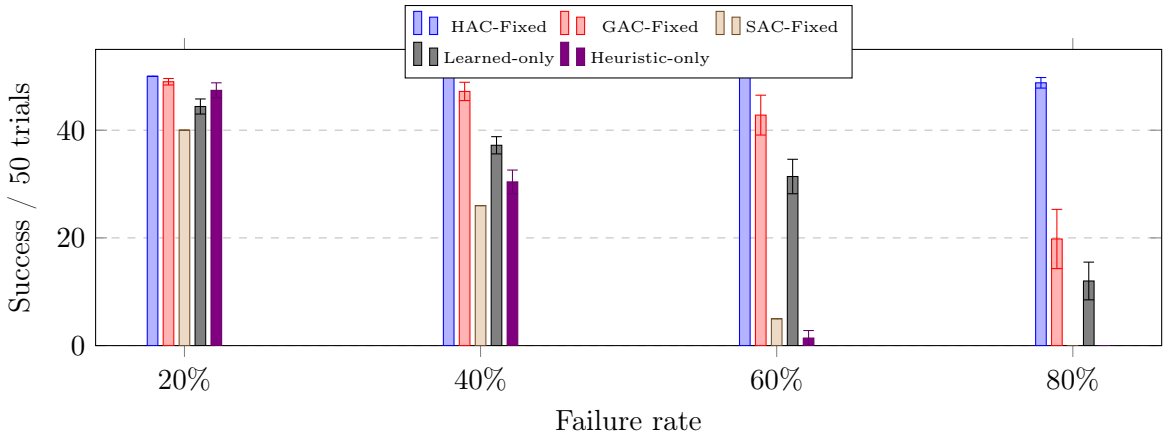


Figure 1: Connectivity recovery success count (out of 50 trials) across node failure rates under random deployment (mean ± std over 5 seeds). HAC-Fixed sustains near-complete recovery with minimal cross-seed variance even at 80% node loss.

coverage (≥ 0.988), with HAC-Fixed achieving faster recovery under uniform deployment (REC of 0.003 vs. 0.037 at 20% failure and 0.037 vs. 0.051 at 40% failure). Under the most severe 80% failure regime, R2C's reported coverage exceeds that of HAC-Fixed by an absolute margin of 7.7 pp in random deployment (0.681 vs. 0.604) and 6.2 pp in uniform deployment (0.668 vs. 0.606), meaning that HAC-Fixed retains 88.7% (random) and 90.7% (uniform) of R2C's coverage. Considering that R2C relies on Quickhull boundary extraction, multipartite cross-subnetwork edge construction, and an 8-layer, 512-dimensional spectral GCN, whereas HAC-Fixed uses a single global 4-layer, 128-dimensional GCN with constant 50/50 fusion weights, these results confirm that the primary resilience gain stems from coupling a local virtual-force prior with a global learned policy rather than complex subnetwork partitioning.

8.3 Topology Quality and Anti-Clustering Analysis

HAC-Fixed completely eliminates the GNN spatial-collapse failure mode while sustaining near-100% topological reconnection. Figure 2 plots the post-recovery average node degree (AND) across failure rates under random deployment. Whereas GAC-Fixed and Learned-only exhibit inflated node degrees (AND of 13.68 and 10.24 at 20% failure) caused by inward GNN aggregation bias, HAC-Fixed tracks the low, smoothly decreasing degree profile of Heuristic-only and SAC-Fixed (AND declining from 6.72 at 20% failure to 2.58 at 80% failure). Thus, velocity-level fusion with the buffered virtual-force field preserves uniform spatial dispersion without sacrificing global reconnection reach.

9 Ablation Studies

9.1 Synergistic Branch Contribution

Velocity-level hybrid fusion exhibits strict super-additive synergy under severe fragmentation rather than merely selecting the better constituent branch. Comparing HAC-Fixed against its isolated branches in Tables 3 and 4 rules out simple branch dominance. Under random deployment at 80% failure, Heuristic-only succeeds in 0.0/50 trials and Learned-only succeeds in 12.0/50 trials, yet their 50/50 velocity fusion achieves 48.8/50 successes (exceeding the

Table 4: Main benchmark results under uniform deployment (mean ± std over 5 seeds, 50 trials per seed).

Architecture	Failure	Success/50 ↑	REC ↓	COV ↑
HAC-Fixed	20%	50.0 ± 0.0	0.003 ± 0.000	1.000 ± 0.000
	40%	50.0 ± 0.0	0.037 ± 0.002	0.998 ± 0.000
	60%	50.0 ± 0.0	0.258 ± 0.012	0.957 ± 0.003
	80%	$\mathbf{49.0 \pm 2.0}$	$\mathbf{0.641 \pm 0.052}$	$\mathbf{0.606 \pm 0.005}$
GAC-Fixed	20%	50.0 ± 0.0	0.011 ± 0.003	0.992 ± 0.003
	40%	50.0 ± 0.0	0.200 ± 0.014	0.843 ± 0.019
	60%	47.0 ± 2.8	0.574 ± 0.013	0.478 ± 0.023
	80%	25.8 ± 6.1	0.772 ± 0.057	0.300 ± 0.012
SAC-Fixed	20%	49.8 ± 0.4	0.016 ± 0.006	0.996 ± 0.002
	40%	35.2 ± 3.7	0.256 ± 0.060	0.944 ± 0.017
	60–80%	0.0 ± 0.0	n/a	n/a
Learned-only	20%	50.0 ± 0.0	0.008 ± 0.003	0.998 ± 0.002
	40%	49.2 ± 0.7	0.228 ± 0.022	0.880 ± 0.021
	60%	33.8 ± 6.4	0.625 ± 0.012	0.581 ± 0.032
	80%	11.6 ± 6.5	0.866 ± 0.070	0.337 ± 0.024
Heuristic-only	20%	49.8 ± 0.4	0.006 ± 0.002	1.000 ± 0.000
	40%	44.0 ± 1.3	0.077 ± 0.017	0.998 ± 0.000
	60%	0.8 ± 0.7	0.110 ± 0.081	0.977 ± 0.009
	80%	0.0 ± 0.0	n/a	n/a

Table 5: Comparison between HAC-Fixed and published numbers of R2C (Peng et al., 2026). Lower REC ↓ and higher COV ↑ indicate superior recovery speed and spatial coverage.

	Random Deployment				Uniform Deployment			
Failure	R2C (Peng et al., 2026)		HAC-Fixed		R2C (Peng et al., 2026)		HAC-Fixed	
Rate	REC↓	COV↑	REC↓	COV↑	REC↓	COV↑	REC↓	COV↑
20%	0.023	0.996	0.061	0.997	0.037	1.000	0.003	1.000
40%	0.070	0.989	0.136	0.988	0.051	0.999	0.037	0.998
60%	0.204	0.951	0.307	0.923	0.181	0.978	0.258	0.957
80%	0.625	**0.681**	0.666	**0.604**	0.617	**0.668**	0.641	**0.606**

sum of both constituent branches by 36.8 trials). The heuristic branch prevents early local link breakage and spatial collapse (Proposition 1), creating well-spaced local clusters that the global GCN branch then steers into contact across long-range voids.

9.2 Adaptive Fusion Schedule Steepness Sweep

Steep adaptive gating schedules degrade recovery robustness under 80% node failure, whereas smoothing the transition slope restores near-optimal performance. As shown in Table 6, transplanting the steep schedule $k = 30, s_0 = 0.125$ into HAC-Adaptive reduces 80%-failure recovery success to 20/50 (versus 48.8/50 for HAC-Fixed). Shifting the transition midpoint later to $s_0 = 0.5$ while keeping the steep slope $k = 30$ further degrades success to 8/50, proving that transition timing alone is not the remedy. Conversely, progressively smoothing the slope to $k = 10, s_0 = 0.5$ (34/50) and $k = 5, s_0 = 0.7$ (48/50) recovers performance to match HAC-Fixed while achieving slightly faster recovery (REC = 0.652), higher coverage (COV = 0.619, +1.5 pp), and lower clustering (AND = 2.46) as detailed in Table 7. Cross-deployment testing on the uniform layout confirms this trend, where smoothing from $(30, 0.125)$ to $(5, 0.7)$ boosts 80%-failure recovery

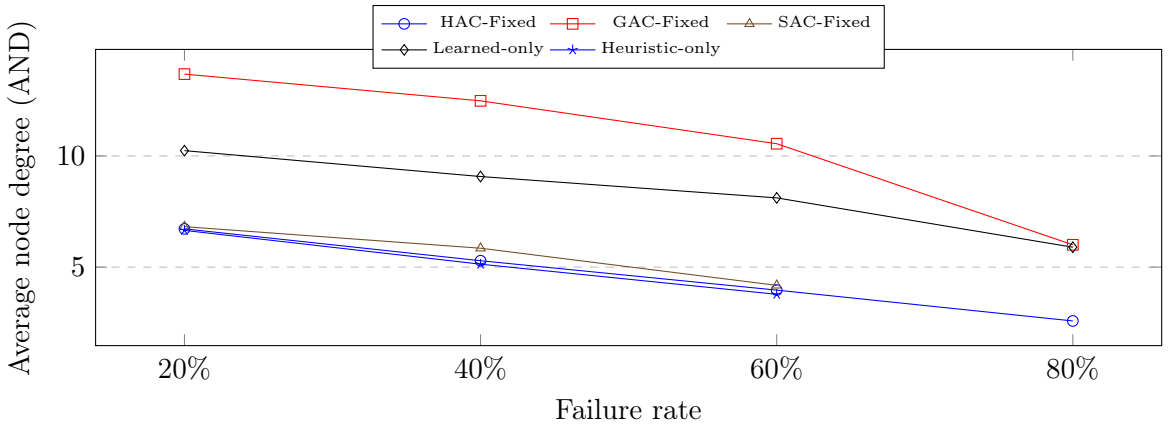


Figure 2: Average node degree (AND) across failure rates under random deployment. HAC-Fixed avoids the high-degree spatial collapse of GAC-Fixed and Learned-only while maintaining the highest recovery success rate.

Table 6: Adaptive fusion schedule sweep across slope k and midpoint s_0 under random deployment (Success/50 across failure rates). Widening the transition window ($k = 5, s_0 = 0.7$) restores robustness at 80% failure.

Schedule (k, s_0)	20%	40%	60%	80%
Steep Early $(30, 0.125)$	50	50	45	20
Moderate Early $(10, 0.125)$	50	50	46	17
Steep Mid $(30, 0.5)$	50	50	43	8
Moderate Mid $(10, 0.5)$	50	50	49	34
Smooth Late $(5, 0.7)$	**50**	**50**	**50**	**48**

from 7/50 to 37/50.

9.3 Granularity Substitution Counter-Experiment (HSC)

Replacing HAC's node-level global GCN with a subnetwork-level rigid-translation GCN (HSC) causes total recovery collapse at 80% **failure that schedule smoothing cannot rescue.** To test whether smooth scheduling can compensate for restricted spatial expressiveness, we evaluate HSC, which fuses the heuristic virtual-force branch with SAC's subnetwork-level rigid translation policy via Eq. (4). Table 8 shows that while schedule smoothing $(5, 0.7)$ improves HSC's recovery at 60% failure from 16/50 to 33/50 (random) and from 13/50 to 40/50 (uniform), all HSC variants collapse to 0/50 or 1/50 at 80% failure. This counter-experiment cleanly separates scheduling instability from architectural expressiveness: when a policy's output space is restricted to rigid subnetwork translation, no curriculum schedule can supply the node-level degrees of freedom required under extreme fragmentation.

9.4 Initialization Robustness and Variance Suppression

Coupling the learned policy with a parameter-free virtual-force heuristic suppresses cross-seed initialization variance by a factor of three to five under extreme failure. Purely learned graph policies exhibit high sensitivity to random weight initialization when evaluated

Table 7: Detailed comparison between HAC-FIXED and smooth adaptive scheduling HAC ($k = 5, s_0 = 0.7$) at 80% failure under random deployment.

Variant	REC ↓	COV ↑	AND	Success/50 ↑
HAC-FIXED	0.666 ± 0.032	0.604 ± 0.005	2.581 ± 0.023	48.8 ± 0.98
HAC-SMOOTH $(5, 0.7)$	0.652	0.619	2.460	48.0

Table 8: Granularity substitution counter-experiment with HSC (heuristic branch fused with subnetwork-level rigid translation, Success/50). Unlike node-level HAC, HSC collapses at 80% failure regardless of schedule smoothness.

	Random Deployment		Uniform Deployment	
Variant	60%	80%	60%	80%
HSC-FIXED	40	0	42	0
HSC-ADAPTIVE $(30, 0.125)$	16	0	13	0
HSC-SMOOTH $(5, 0.7)$	33	1	40	0

on multi-step open-loop rollouts: across five properly locked random seeds at 80% failure, GAC-FIXED and LEARNED-ONLY display large success-rate standard deviations of ± 5.5 and ± 3.5 trials in random deployment (and ± 6.1 and ± 6.5 in uniform deployment). Moreover, an unseeded single run of GAC-FIXED at 20% failure yielded an optimistic outlier (COV $= 0.830$, AND $= 11.87$) compared to the true five-seed distribution (COV $= 0.747 \pm 0.009$, AND $= 13.68 \pm 0.28$). In contrast, HAC-FIXED anchors every rollout step to the deterministic physical force field in Eq. (2), compressing the 80%-failure success standard deviation to ± 0.98 (random) and ± 2.0 (uniform) while holding coverage standard deviation to ± 0.005.

10 Discussion

10.1 Trajectory-Level Complementarity and Scheduling Guidelines

Simple velocity-level hybrids succeed because local virtual forces and global graph convolutions resolve non-overlapping failure modes at every step of an individual trajectory. The virtual-force branch guarantees local link retention and collision avoidance within connected components (Proposition 1) but is blind to components beyond R_c. Conversely, the global k-NN GCN perceives the relative geometry of disconnected subnetworks across the entire arena but tends to over-cluster nodes if left unchecked. Fusing both branches at the velocity level allows the heuristic term to continuously regularize intra-component spacing while the GCN steers component centroids into reconnection range.

Practitioners designing adaptive hybrid controllers should verify output-space expressiveness before tuning handover schedules, and default to constant or wide-transition schedules. Comparing HAC and HSC yields a clear engineering principle: adaptive schedule tuning is only meaningful when the learned branch possesses node-level actuation authority. Furthermore, because steep temporal handovers degrade end-to-end training by inflating autonomy exposure, zero-tuning constant fusion (50/50) or wide-window smooth scheduling ($k = 5$) provides a much safer default than sharp stage switches.

10.2 Limitations and Future Work

Closing the remaining 6.2 to 7.7 pp coverage gap under 80% failure and extending differentiable rollout horizons represent promising avenues for future research. First, while HAC-Fixed retains roughly 90% of R2C's reported coverage under 80% node loss without boundary-hull machinery, introducing lightweight node-level boundary indicators or increasing GCN capacity from 4 layers (128 hidden units) toward deeper spectral architectures could close the remaining 6.2 to 7.7 pp margin without sacrificing node-level simplicity. Second, due to backpropagation-through-time memory constraints, our models were trained on 40-step rollouts and evaluated over 400-step horizons; curriculum-extended rollout horizons or truncated gradient windowing could further improve late-stage trajectory efficiency. Finally, conducting multi-seed sweeps across iso-contours of autonomy exposure $\mathrm{AE}(k, s_0; 0.5)$ will directly test the theoretical invariance predicted in Section 6.3.

11 Conclusion

This paper systematically disentangled the roles of decision paradigm, spatial reasoning granularity, and temporal fusion scheduling in self-healing UAV swarm connectivity recovery. By introducing Hybrid Adaptive Convolution (HAC), we demonstrated that fusing a parameter-free buffered virtual-force heuristic with a single global node-level GCN under a constant 50/50 weighting outperforms purely heuristic, purely learned, and subnetwork-level rigid baselines across random and uniform deployments, retaining 88.7% to 90.7% of the coverage reported by complex multipartite architectures under 80% node failure without boundary-hull extraction. Furthermore, we showed theoretically and empirically that steep adaptive fusion schedules impair end-to-end learning by prematurely exposing the learned branch to high autonomy, whereas smooth transition schedules restore robustness for node-level policies while failing to rescue expressively constrained subnetwork-level rigid controllers. Together with our cardinality-invariant connectivity loss formulation and open-source benchmark suite, these findings provide a rigorous foundation for designing simple, resilient hybrid swarm controllers.

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A Implementation Details and Methodological Verification Log

To ensure full reproducibility and clarify experimental design choices relative to prior literature (Peng et al., 2026), we document all simulation specifications and numerical verifications below:

- **Failure-node sampling:** Failed UAVs at ratios $\{20\%, 40\%, 60\%, 80\%\}$ are sampled uniformly at random without replacement from the initial deployed swarm.
- **Cardinality-invariant connectivity loss verification:** As derived in Eq. (5), omitting the $-\log(n_i n_j)$ normalization term in a log-sum-exp soft-minimum surrogate introduces a severe component-size bias. On a controlled synthetic benchmark featuring two disconnected components separated by an identical true minimum distance (≈ 100 to $108\,\mathrm{m}$) under two cardinality configurations (3×3 nodes vs. 30×30 nodes), the unnormalized log-sum-exp estimator returned 2.85 for the 3×3 pair and -230.58 for the 30×30 pair. In contrast, our cardinality-invariant estimator in Eq. (5) returned 112.71 and 109.54, respectively, accurately tracking the true geometric separation regardless of cluster size.
- **Boundary buffer specification:** The arena boundary repulsion buffer is set to $d_e = 0.15R_c = 18\,\mathrm{m}$, matching the scale of the inter-UAV equilibrium band $d_u - d_l = 0.1R_c = 12\,\mathrm{m}$.
- **Differentiable rollout horizon:** Training uses $T_{\text{train}} = 40$ differentiable rollout steps per epoch for backpropagation-through-time tractability, while evaluation runs for the full horizon $T = 400$ steps.
- **Grid-based coverage evaluation:** Swarm coverage $C_{\text{cov}}(t)$ is evaluated over a deterministic uniform grid at $5\,\mathrm{m}$ spatial resolution across the $1000 \times 1000\,\mathrm{m}^2$ arena.
- **Full-pipeline random seeding:** Every multi-seed experiment locks Python, NumPy, CPU PyTorch (`torch.manual_seed`), and CUDA PyTorch (`torch.cuda.manual_seed_all`) generators prior to both network initialization and environment reset.

B Full Hyperparameter Table

Table 9: Hyperparameters shared across all evaluated architectures and training runs.

Parameter	Value
Graph convolution layers L	4
Hidden feature dimension	128
Global k-NN degree k	8
Optimizer	Adam
Learning rate	1×10^{-3}
Gradient norm clipping threshold	1.0
Training epochs	300
Training rollout length T_{train}	40 steps
Evaluation horizon T	400 steps
Movement cost weight λ	0.01
Heuristic decay rates α, β, δ	0.01
Equilibrium buffer zone $[d_l, d_u]$	$[0.8R_c, 0.9R_c]$
Communication and coverage radii $R_c = R_{\text{cov}}$	120 m
Maximum UAV speed $\hat{v}_{\max}$	1 m/s
Mission area $\mathcal{A}$	$1000 \times 1000\,\text{m}^2$
Random deployment initial UAV count	200
Uniform deployment initial UAV count	196 (14×14 grid)

C Additional Topology Metrics for Uniform Deployment

Tables 3 and 4 in the main text report full Success/50, REC, and COV statistics across all failure ratios. For completeness, the post-recovery average node degree (AND) trajectories under uniform deployment across failure ratios (20% → 40% → 60% → 80%) are as follows:

- **HAC-Fixed:** 5.72 → 4.32 → 3.30 → 2.37
- **GAC-Fixed:** 5.88 → 6.86 → 8.75 → 5.81
- **SAC-Fixed:** 5.74 → 4.46 → n/a → n/a
- **Learned-only:** 5.76 → 6.37 → 7.34 → 5.29
- **Heuristic-only:** 5.72 → 4.36 → 2.87 → n/a

Consistent with the random-deployment analysis in Figure 2, HAC-Fixed maintains a low, monotonically decreasing node degree across all failure regimes under uniform deployment, whereas GAC-Fixed and Learned-only exhibit pronounced mid-regime clustering (8.75 and 7.34 at 60% failure).

D Code and Reproducibility Artifacts

The companion repository contains the complete simulation environment (`code/sim/`), heuristic virtual-force and global GCN model definitions (`code/models/`), differentiable rollout training and evaluation scripts (`code/scripts/`), multi-seed execution shell scripts, and aggregated CSV logs (`code/results/`) required to reproduce the tables and figures in this manuscript.